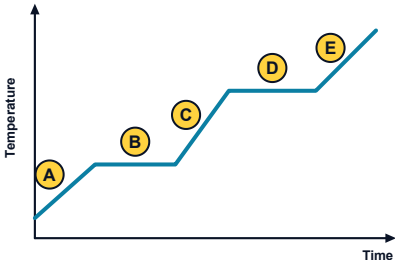
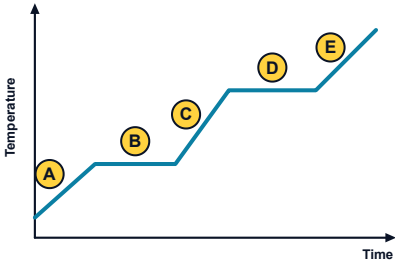
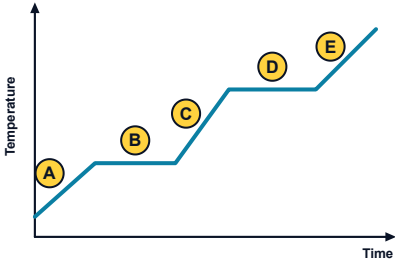
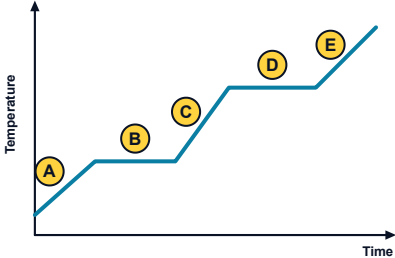
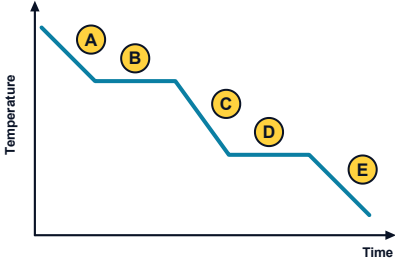


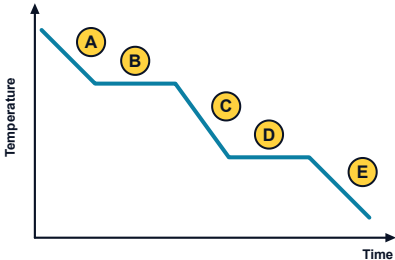
Changes of state and specific latent heat

Particle model of matter
Paper 1 · Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	State the standard unit of specific latent heat.	Joule per kilogram (J/kg)	☐
2 ★	Define latent heat.	The energy needed to change the state of a substance with no change in temperature.	☐
3 ★	Define specific latent heat.	The energy needed to change the state of 1 kg of a substance with no change in temperature.	☐
4	Define specific latent heat of fusion.	The energy needed to change 1 kg of a substance from solid to liquid with no change in temperature.	☐
5	Define specific latent heat of vaporisation.	The energy needed to change 1 kg of a substance from liquid to vapour (gas) with no change in temperature.	☐
6	Which letter on the heating curve shows the substance melting? 	B	☐

QUESTION	ANSWER	✓
7 Which letter on the heating curve shows the substance boiling? 	D	☐
8 Which letter on the heating curve shows a solid getting hotter? 	A	☐
9 Which letter on the heating curve shows a liquid getting hotter? 	C	☐
10 Which letter on the cooling curve shows the substance freezing? 	D	☐

QUESTION	ANSWER	✓
11 Which letter on the cooling curve shows a liquid cooling? 	C	☐
12 State the equation linking energy for a change of state, mass and specific latent heat.	energy for a change of state = mass × specific latent heat ($E = m L$)	☐
13 ★ Explain why the temperature stays constant while a substance melts.	The energy supplied increases the potential energy of the particles, not their kinetic energy.	☐
14 State the difference between specific heat capacity and specific latent heat.	Specific heat capacity: energy to change the temperature of 1 kg by 1 °C. Specific latent heat: energy to change the state of 1 kg.	☐